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FIXING STRUCTURE FOR PIPE NUT

Abstract

A fixing structure includes an insertion hole, a slit, a first stopper, and a second stopper. The insertion hole is provided in a vehicle body frame. A pipe nut main body is inserted through the insertion hole so as to be rotatable. The slit is provided in the vehicle body frame and extends from the insertion hole. The first stopper is provided at a base end of the pipe nut main body and abuts on an outer surface of the vehicle body frame to restrict movement of the pipe nut main body in a longitudinal direction. The second stopper extends from a side surface of the pipe nut main body, passes through the slit, and is disposed inside the vehicle body frame. The second stopper is joined to an inner surface of the vehicle body frame at a position rotated around the central axis from the slit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority from Japanese Patent Application No. 2024-032377 filed on Mar. 4, 2024, the entire contents of which are hereby incorporated by reference.

BACKGROUND

[0002] The disclosure relates to a fixing structure for a pipe nut provided on a vehicle body frame for fastening a sub frame to the vehicle body frame by a bolt.

[0003] Conventionally, front side frames extending in a front-rear direction are disposed as vehicle body frames on both sides of a front portion of a vehicle body of an automobile or the like, for example. On lower portions of the front side frames, for example, a suspension cross member that supports suspensions and the like is extended in the vehicle width direction as a sub frame.

[0004] Both ends of such a suspension cross member are generally attached to each of the front side frames by bolt fastening. Therefore, in order to fasten both ends of the suspension cross member to each of the front side frames by bolts, a pipe nut having a female screw portion therein is joined to the front side frame.

[0005] As such a fixing structure for increasing the attachment rigidity of the pipe nut, various techniques have been conventionally proposed.

[0006] For example, Japanese Unexamined Patent Application Publication No. 2011-230602 discloses a structure in which a reinforcing member for suppressing tilting of a pipe nut is provided, and the reinforcing member is joined to an inner surface of a vehicle body frame to fix the pipe nut. Such a reinforcing member can also increase the rigidity of the vehicle body frame, in addition to increasing the attachment rigidity of the pipe nut.

[0007] In the case of such a fixing structure, it is common to weld the reinforcing member to the inner surface of the vehicle body frame in a state where the pipe nut and the reinforcing member are sub-assembled.

[0008] In order to further improve the attachment rigidity with respect to such a fixing structure in which welding is performed just from an inside of the vehicle body frame, it is preferable to perform welding also from an outside of the vehicle body frame.

[0009] As a fixing structure for improving the attachment rigidity of such a pipe nut, a fixing structure is also conceivable in which a flange to be welded in contact with an outer surface of the vehicle body frame is provided at a lower end of the pipe nut.

SUMMARY

[0010] An aspect of the disclosure provides a fixing structure. The fixing structure is for a pipe nut that fixes a pipe nut main body to a vehicle body frame having a hollow cross section. The fixing structure includes an insertion hole, a slit, a first stopper, and a second stopper. The insertion hole is provided in the vehicle body frame and through which the pipe nut main body is inserted in such a manner that the pipe nut main body is rotatable around a central axis of the pipe nut main body. The slit is provided in the vehicle body frame and extends from the insertion hole. The first stopper is provided at a base end of the pipe nut main body and abuts on an outer surface of the vehicle body frame to restrict movement of the pipe nut main body in a longitudinal direction. The second stopper extends from a side surface of the pipe nut main body, passes through the slit, and is disposed inside the vehicle body frame. The second stopper is joined to an inner surface of the vehicle body frame at a position rotated around the central axis from the slit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments and, together with the specification, serve to describe the principles of the disclosure.

[0012] FIG. 1 is an exploded perspective view of a vehicle body frame and a suspension cross member according to an embodiment;

[0013] FIG. 2 is an enlarged perspective view of a main part of the vehicle body frame according to the embodiment;

[0014] FIG. 3 is an enlarged exploded perspective view of the main part of the vehicle body frame according to the embodiment;

[0015] FIG. 4 is a cross-sectional view taken along line IV-IV of FIG. 2 according to the embodiment;

[0016] FIG. 5 is a perspective view of a pipe nut according to the first embodiment from an obliquely lower side;

[0017] FIG. 6 is a cross-sectional view taken along line VI-VI of FIG. 4 according to the embodiment;

[0018] FIG. 7 is a cross-sectional view corresponding to FIG. 4 according to an embodiment;

[0019] FIG. 8 is an enlarged exploded perspective view of a main part of a vehicle body frame according to an embodiment; and

[0020] FIG. 9 is a perspective view of an inner frame according to the embodiment from an obliquely lower side.

DETAILED DESCRIPTION

[0021] In the fixing structure, it may be difficult to sub-assemble the pipe nut and the reinforcing member. In this case, assemblability may be deteriorated.

[0022] It is desirable to provide a fixing structure for a pipe nut capable of ensuring the attachment rigidity of the pipe nut with respect to a vehicle body frame, enhancing assemblability, and further enhancing the rigidity of the vehicle body frame.

[0023] Hereinafter, embodiments of the disclosure will be described with reference to the drawings.

[0024] Note that the following description is directed to illustrative examples of the disclosure and not to be construed as limiting to the disclosure. Factors including, without limitation, numerical values, shapes, materials, components, positions of the components, and how the components are coupled to each other are illustrative only and not to be construed as limiting to the disclosure.

Further, elements in the following example embodiments which are not recited in a most-generic independent claim of the disclosure are optional and may be provided on an as-needed basis. The drawings are schematic and are not intended to be drawn to scale. Throughout the present specification and the drawings, elements having substantially the same function and configuration are denoted with the same numerals to avoid any redundant description.

[0025] Further, in the following description, in a case where “joining” is described, the joining method is performed using a joining means represented by fusion welding, mechanical joining, or the like. Note that, in the drawings, a spot welding symbol is attached to a spot welding point, and display of the welding symbol is omitted in other welding points.

[0026] As illustrated in FIG. 1, an engine room 2 is provided in a front portion of the vehicle body 1.

[0027] The engine room 2 is partitioned from the rear cabin 3 by a toe board 4.

[0028] As frames constituting such an engine room 2, the vehicle body 1 includes a pair of left and right front side frames 6, a pair of left and right upper side frames 7, and a suspension cross member 9. These frames are formed by, for example, pressing a sheet metal member made of a

high-strength steel plate or the like.

[0029] Note that the basic structure around the pair of left and right front side frames **6** and the pair of left and right upper side frames **7** is bilaterally symmetrical. Therefore, the left side of the vehicle body will be described below as an example.

[0030] As illustrated in FIG. **1**, the front side frames **6** extend from a lower portion of the toe board **4** toward the front of the vehicle body **1** on outsides of the toe board **4** in a vehicle width direction.

[0031] The upper side frames **7** are provided above the front side frames **6** and outside the front side frames **6** in the vehicle width direction. The upper side frames **7** extend forward of the vehicle body along the engine room **2**.

[0032] A suspension tower **8** is provided between each of the front side frames **6** and each of the upper side frames **7** near the rear of the engine room **2**.

[0033] The suspension tower **8** houses a suspension device (not illustrated) therein. The suspension tower **8** supports an upper portion of the suspension device.

[0034] Furthermore, a wheel apron **5** is provided between each of the front side frames **6** and each of the upper side frames **7** in front of the suspension tower **8**.

[0035] The suspension cross member **9** is disposed as a sub frame on lower portions of the front side frames **6**. That is, the suspension cross member **9** extends in the vehicle width direction at the lower portions of the front side frames **6**. The suspension cross member **9** has a shape in which a central portion is curved downward.

[0036] Ends of the suspension cross member **9** in the vehicle width direction are joined to bottom surfaces of the front side frames **6** by bolt fastening.

[0037] In order to fasten such a suspension cross member **9** by bolts, the front side frames **6** as a vehicle body frame are each provided with a pair of front and rear pipe nuts **14**.

[0038] The configurations of the front side frames **6** and the suspension cross member **9** will be described in detail below.

[0039] As illustrated in FIGS. **1** and **2**, the front side frames **6** each include an inner frame **12** and an outer frame **13**. Note that, in FIG. **1**, a part of the outer frame **13** of the left front side frame **6** is omitted.

[0040] As illustrated in FIGS. **2** to **4**, the inner frame **12** has a substantially hat-shaped cross section protruding inward in the vehicle width direction. Such an inner frame **12** includes an inner wall **12a**, an upper wall **12b**, a lower wall **12c**, and a pair of upper and lower inner flanges **12d**.

[0041] The inner wall **12a** extends in the vehicle body vertical direction.

[0042] The upper wall **12b** and the lower wall **12c** extend outward in the vehicle width direction from both ends of the inner wall **12a**.

[0043] The pair of upper and lower inner flanges **12d** protrudes in the vehicle body up-down direction from an outer surface of the inner frame **12**. For example, the upper wall **12b** and the lower wall **12c** extend in the vehicle body up-down direction from outer edges in the vehicle width direction. Each of the inner flanges **12d** functions as a flange for joining the inner frame **12** to the outer frame **13**. Further, the inner flange **12d** extending from the lower wall **12c** also serves as an abutting portion of a pair of rotation restricting portions **14h** described later.

[0044] As illustrated in FIGS. **2** to **4**, the outer frame **13** is disposed at a position facing the outside of the inner frame **12** in the vehicle width direction. The outer frame **13** is a flat plate that closes the opening of the inner frame **12**.

[0045] Therefore, the outer frame **13** has a pair of upper and lower outer flanges **13a** at positions facing the inner flanges **12d**.

[0046] Such an outer frame **13** is joined in a state where the outer flanges **13a** are made to abut on the inner flanges **12d**.

[0047] Thus, as illustrated in FIG. **4**, the front side frame **6** has a substantially rectangular cross section having a hollow cross section **15**.

[0048] With respect to the front side frame **6**, the pair of front and rear pipe nuts **14** can be inserted

into the lower wall **12c**. Therefore, the lower wall **12c** is provided with insertion holes for inserting the respective pipe nuts **14**.

[0049] For example, as illustrated in FIG. 3, the lower wall **12c** is provided with a pair of front and rear lower insertion holes **12e**. Furthermore, the lower wall **12c** is provided with a pair of front and rear slits **12f** extending from each of the lower insertion holes **12e** in the front-rear direction of the vehicle body. Note that the slits **12f** are provided for inserting a pair of reinforcing plates **14c** to be described later.

[0050] As illustrated in FIG. 5, the pipe nut **14** inserted into each set of the lower insertion holes **12e** and the slits **12f** includes a pipe nut main body **14a**, joining flanges **14b**, and a pair of reinforcing plates **14c**.

[0051] As illustrated in FIGS. 4 and 6, the pipe nut main body **14a** has, for example, a substantially cylindrical shape with one end closed. The hole formed in the pipe nut main body **14a** is set as a bolt insertion hole **14e**. Furthermore, a female screw portion **14f** is formed on an inner peripheral surface of the bolt insertion hole **14e**. Note that a bolt **19** to be described later can be screwed into the female screw portion **14f**. The outer diameter of the pipe nut main body **14a** is set slightly smaller than the inner diameter of the lower insertion hole **12e**. Thus, the pipe nut main body **14a** can be inserted into the lower insertion hole **12e**. Furthermore, the pipe nut main body **14a** inserted into the lower insertion hole **12e** is rotatable around a central axis **0** of the pipe nut main body **14a**.

[0052] The joining flange **14b** is provided at a base end of the pipe nut main body **14a**. As illustrated in FIG. 5, the basic shape of the joining flange **14b** is, for example, a rectangular flat plate shape. Such a joining flange **14b** extends in a direction orthogonal to the central axis **O** of the pipe nut main body **14a**. In one embodiment, the joining flange **14b** serves as a “first stopper” that restricts movement of the pipe nut main body **14a** in a longitudinal direction when abutting on the outer surface of the lower wall **12c**.

[0053] Furthermore, the extended portion of the joining flange **14b** serves as a welded portion with the lower wall **12c**. That is, the joining flange **14b** allows spot-welding of four corners of the joining flange **14b** in a state of being in contact with the outer surface of the lower wall **12c**.

[0054] Note that, in the following description, for convenience, as illustrated in FIG. 5, a direction orthogonal to the central axis **O** and in which the long side of the joining flange **14b** extends is defined as a first direction **D1**. Further, a direction orthogonal to the central axis **O** and orthogonal to the first direction **D1** is defined as a second direction **D2**.

[0055] Such a joining flange **14b** is provided with rotation restricting portions **14h** protruding from the pair of long sides in the second direction **D2**. The protrusion amount of each rotation restricting portion **14h** in the second direction **D2** is set based on the distance from the lower insertion hole **12e** of the front side frame **6** to the inner flange **12d**. For example, as illustrated in FIG. 3, when the pipe nut main body **14a** is inserted into the lower insertion hole **12e** and the joining flange **14b** is disposed so that the first direction **D1** coincides with a longitudinal direction of the front side frame **6**, any one of the pair of rotation restricting portions **14h** is set to abut on the inner flange **12d**. Thus, each rotation restricting portion **14h** can restrict rotation of the pipe nut main body **14a** inserted into the lower insertion hole **12e**.

[0056] Further, as illustrated in FIG. 5, a boss portion **14d** protruding downward is formed at a lower end of the joining flange **14b**.

[0057] The boss portion **14d** is formed in an annular shape. A bearing surface **14g** is provided on a protrusion surface of the boss portion **14d**.

[0058] As illustrated in FIGS. 4 and 6, an upper surface of the suspension cross member **9** can abut on the bearing surface **14g**. That is, the bearing surface **14g** functions as an attachment surface when the suspension cross member **9** is fasten by bolts. Therefore, the bearing surface **14g** is substantially flat.

[0059] As illustrated in FIG. 5, each of the pair of reinforcing plates **14c** is formed by, for example, a flat plate having a substantially rectangular shape in front view. Each reinforcing plate **14c** is

provided on a side surface of the pipe nut main body **14a**. These reinforcing plates **14c** are provided at positions that are rotationally symmetric with respect to the central axis O of the pipe nut main body **14a**. Furthermore, on the side surface of the pipe nut main body **14a**, each of the reinforcing plates **14c** is disposed so that a plate surface extends in the central axis O direction and the second direction D2.

[0060] A height dimension H1 of each of the reinforcing plates **14c** is set to be substantially the same as a height dimension H2 in an internal space of the front side frame **6** in the direction of the central axis O of the pipe nut main body **14a** (see FIGS. 4 and 5).

[0061] Further, in the second direction D2, a width dimension W1 from an extending end of one of the reinforcing plates **14c** to an extending end of the other of the reinforcing plates **14c** is set substantially equal to a width dimension W2 in the vehicle width direction in the internal space of the front side frame **6** (see FIGS. 4 and 5).

[0062] Here, a gap wider than at least the plate thickness of the lower wall **12c** is provided between the base end of each of the reinforcing plates **14c** and the joining flange **14b**. Further, the dimension of each slit **12f** in the front-rear direction of the vehicle body through which each reinforcing plate **14c** is inserted is set slightly larger than the dimension of each reinforcing plate **14c** in the second direction D2. Furthermore, the size of each slit **12f** in the vehicle width direction through which each reinforcing plate **14c** is inserted is set to be slightly larger than the thickness of each reinforcing plate **14c**.

[0063] Thus, each reinforcing plate **14c** can be inserted into each slit **12f**. Furthermore, each of the reinforcing plates **14c** inserted into each of the slits **12f** is disposed inside the front side frame **6** and is rotatable around the central axis O of the pipe nut main body **14a**.

[0064] Next, a procedure for assembling such a pipe nut **14** to the front side frame **6** will be described. The pipe nut **14** is assembled to the inner frame **12** before the outer frame **13** is welded.

[0065] First, as illustrated in FIG. 3, an assembly worker (hereinafter, the worker) aligns the pipe nut **14** below the inner frame **12** so that the central axis O of the pipe nut main body **14a** coincides with the center of the lower insertion hole **12e**. At this time, the worker performs adjustment so that the first direction D1 of the pipe nut **14** coincides with the vehicle width direction of the inner frame **12**.

[0066] Thus, each of the reinforcing plates **14c** is positioned at a position where the reinforcing plate can be inserted into each of the slits **12f**.

[0067] In such a state, the worker inserts the pipe nut main body **14a** and the reinforcing plates **14c** into the lower insertion hole **12e** and the slits **12f**.

[0068] Then, when the joining flange **14b** abuts on the outer surface of the lower wall **12c**, the worker rotates the joining flange **14b**. The joining flange **14b** is rotated to a position where any one of the rotation restricting portions **14h** abuts on the inner flange **12d**. That is, the inner flange **12d** restricts rotation of the pipe nut **14** by abutting on the rotation restricting portion **14h**. In one embodiment, the inner flange **12d** may serve as a “first protrusion” for restricting the rotation of the pipe nut **14**.

[0069] Thus, the first direction D1 of the pipe nut **14** coincides with the front-rear direction of the vehicle body of the inner frame **12**. That is, when the rotation restricting portion **14h** is made to abut on the inner flange **12d**, each of the reinforcing plates **14c** extends in a direction orthogonal to the longitudinal direction of the inner frame **12**.

[0070] At this time, an edge on a base end side of each of the reinforcing plates **14c** is made to abut on an inner surface of the lower wall **12c**. Therefore, the pipe nut **14** is prevented from falling off from the inner frame **12**.

[0071] Furthermore, edges of one of the reinforcing plate **14c** abut on inner surfaces of the inner wall **12a** and the upper wall **12b**. Further, an edge of the other of the reinforcing plates **14c** abuts on the inner surface of the upper wall **12b**.

[0072] In such a state, the worker joins the reinforcing plates **14c** to the inner surfaces of the inner

wall **12a** and the upper wall **12b** by arc welding. Furthermore, the worker joins the joining flange **14b** to the outer surface of the lower wall **12c** by spot welding. Thus, the pipe nut **14** is assembled to the inner frame **12**.

[0073] Then, the worker joins the inner flanges **12d** of the inner frame **12** in a state where the outer flanges **13a** of the outer frame **13** are made to abut on the inner flanges.

[0074] Note that some or all of the assembling work can be performed by a machine tool instead of the worker.

[0075] As illustrated in FIGS. **2**, **4**, and **6**, the pipe nut main body **14a** assembled to the inside of the front side frame **6** is joined to inner surfaces of the front side frame **6** with the reinforcing plates **14c** interposed therebetween in a state of being supported by the reinforcing plates **14c**. Thus, each of the reinforcing plates **14c** can restrict movement of the pipe nut main body **14a** in a prying direction (direction inclined with respect to the central axis) and the longitudinal direction. As described above, in the present embodiment, each of the reinforcing plates **14c** corresponds to an example as a second stopper.

[0076] Furthermore, each of the reinforcing plates **14c** extends in a direction orthogonal to the longitudinal direction of the front side frame **6** inside the front side frame **6**. Each of such reinforcing plates **14c** also serves as a reinforcing member for reinforcing the front side frame **6** inside the front side frame **6**.

[0077] As illustrated in FIGS. **1**, **4**, and **6**, the suspension cross member **9** includes an upper panel **40**, a lower panel **41**, and a reinforcement **42**.

[0078] The upper panel **40** has, for example, an open cross-sectional shape whose cross section is open toward the lower side of the vehicle body. As illustrated in FIG. **4**, such an upper panel **40** includes an attachment portion **40a** and a side wall **40b**.

[0079] The attachment portion **40a** is disposed at a position facing the lower wall **12c**. The attachment portion **40a** abuts on the bearing surface **14g** of the pipe nut **14** and functions as an attachment portion for fixing the suspension cross member **9** to the front side frame **6**.

[0080] Thus, the attachment portion **40a** has a pair of front and rear first bolt insertion holes **40c**.

[0081] The pair of front and rear first bolt insertion holes **40c** is provided at positions facing the respective bolt insertion holes **14e**. That is, the first bolt insertion holes **40c** and the bolt insertion holes **14e** are arranged substantially coaxially.

[0082] The side wall **40b** extends downward from an outer edge of the attachment portion **40a** in the vehicle width direction.

[0083] The lower panel **41** is disposed at a position facing the upper panel **40** with a space therebetween. The lower panel **41** is provided with a second bolt insertion hole **41a** at a position facing each of the first bolt insertion holes **40c**. That is, the second bolt insertion holes **41a** and the first bolt insertion holes **40c** are arranged substantially coaxially.

[0084] The outer edge of the lower panel **41** in the vehicle width direction is bent downward. In this manner, a lower end of the side wall **40b** of the upper panel **40** is joined to an outer surface of the bent lower panel **41** in the vehicle width direction.

[0085] By joining the upper panel **40** and the lower panel **41**, an internal space is formed in the suspension cross member **9**.

[0086] The reinforcement **42** is disposed inside the suspension cross member **9**. For example, the reinforcement **42** extends in the vehicle width direction along the upper panel **40**.

[0087] As illustrated in FIG. **4**, the outer end of the reinforcement **42** in the vehicle width direction is in contact with the inner surface of the attachment portion **40a**. Then, an outer end of the reinforcement **42** in the vehicle width direction is joined to the attachment portion **40a**.

[0088] Further, the joint portion of the reinforcement **42** is provided with a third bolt insertion hole **42a** at a position facing each first bolt insertion hole **40c**. That is, each third bolt insertion hole **42a** and each of the first and second bolt insertion holes **40c** and **41a** are arranged substantially coaxially.

[0089] As illustrated in FIGS. 4 and 6, a pair of front and rear spacers **43** is extended between the reinforcement **42** and the lower panel **41**.

[0090] The pair of front and rear spacers **43** has a substantially cylindrical shape. The hole formed in each spacer **43** is set as a fourth bolt insertion hole **43a** penetrating in the vehicle body vertical direction. That is, the fourth bolt insertion hole **43a** of each spacer **43** is disposed substantially coaxially with each of the first to third bolt insertion holes **40c**, **41a**, and **42a**.

[0091] A bolt **19** can be inserted into the first to fourth bolt insertion holes **40c**, **41a**, **42a**, and **43a** arranged in this manner.

[0092] Note that, in the following description, the first to fourth bolt insertion holes **40c**, **41a**, **42a**, and **43a** are collectively referred to as a bolt insertion hole H.

[0093] With such a configuration, the suspension cross member **9** is fixed to the front side frame **6** by bolt fastening.

[0094] For example, when the suspension cross member **9** is fixed to the front side frame **6**, the attachment portion **40a** abuts on each of the bearing surfaces **14g** from below.

[0095] Further, each of the bolt insertion holes H is positioned with respect to the bolt insertion hole **14e** of each of the pipe nuts **14**.

[0096] The bolt **19** is inserted into the bolt insertion hole H and the bolt insertion hole **14e** of each set positioned from below the bolt insertion hole H. Then, each bolt **19** is screwed into the female screw portion **14f** of the pipe nut **14**. Thus, the suspension cross member **9** is fixed to the front side frame **6** by fastening.

[0097] An arm support portion **45** is provided on the outside in the vehicle width direction of the suspension cross member **9** fixed in this manner.

[0098] For example, as illustrated in FIG. 1, the arm support portion **45** protrudes downward from a bottom surface on the outside in the vehicle width direction of the lower panel **41**. A base end of a suspension lower arm **50** is pivotally supported by the arm support portion **45**. Thus, a distal end side of the suspension lower arm **50** is swingable in the vertical direction.

[0099] A lower portion of a suspension device, which is not illustrated, is supported outside the suspension lower arm **50** in the vehicle width direction. Furthermore, a wheel hub, which is not illustrated, for fixing the front wheel is rotatably supported on the outside in the vehicle width direction of the suspension lower arm **50**.

[0100] According to such an embodiment, the fixing structure for the pipe nut **14** is a fixing structure for the pipe nut **14** that fixes the pipe nut main body **14a** to the front side frame **6** having the hollow cross section **15**, and includes the lower insertion hole **12e** provided in the front side frame **6** and through which the pipe nut main body **14a** is inserted in a state of being rotatable around the central axis O of the pipe nut main body **14a**, the slit **12f** provided in the front side frame **6** and extending from the lower insertion hole **12e**, the joining flange **14b** that is provided at the base end of the pipe nut main body **14a** and abuts on an outer surface of the front side frame **6** to restrict movement of the pipe nut main body **14a** in the longitudinal direction, and the reinforcing plate **14c** extending from the side surface of the pipe nut main body **14a** and disposed inside the front side frame **6** through the slit **12f**, and the reinforcing plate **14c** is joined to the inner surface of the front side frame **6** at a position rotated around the central axis O from the slit **12f**. With these configurations, the fixing structure for the pipe nut **14** can ensure the attachment rigidity of the pipe nut **14** with respect to the vehicle body frame, enhance assemblability, and further enhance the rigidity of the vehicle body frame.

[0101] That is, in the present embodiment, the pair of reinforcing plates **14c** disposed inside the front side frame **6** extends from the side surface of the pipe nut main body **14a**. Further, the front side frame **6** is provided with the lower insertion hole **12e** through which the pipe nut main body **14a** is inserted. Furthermore, the front side frame **6** is provided with the pair of front and rear slits **12f** extending from the lower insertion hole **12e**.

[0102] With these configurations, the fixing structure for the pipe nut **14** of the present embodiment

can be inserted into the front side frame **6** even if each reinforcing plate **14c** extends to the pipe nut main body **14a**. Thus, in a state where the pipe nut main body **14a** and the reinforcing plates **14c** are inserted into the front side frame **6**, the reinforcing plates **14c** can be easily disposed inside the front side frame **6** just by rotating the joining flange **14b** around the central axis O of the pipe nut main body **14a**. Thus, the assemblability can be enhanced as compared with a case where the reinforcing member is assembled after the pipe nut main body **14a** is inserted into the front side frame **6**.

[0103] Then, the reinforcing plates **14c** are joined in a state where the edges of the reinforcing plates **14c** are brought into contact with the inner surface of the front side frame **6**.

[0104] For example, the edges of the reinforcing plates **14c** are firmly joined by arc welding in a state of being in contact with inner surfaces of the inner frame **12**. That is, the pipe nut main body **14a** is firmly joined to the inner surfaces of the front side frame **6** with the reinforcing plates **14c** interposed therebetween in a state of being supported by the reinforcing plates **14c**. Thus, the reinforcing plates **14c** can generate a sufficient drag against a load in the prying direction (direction inclined with respect to the central axis) and the longitudinal direction input to the pipe nut main body **14a** via the suspension cross member **9** during traveling. Therefore, the fixing structure for the pipe nut **14** can ensure the attachment rigidity of the pipe nut **14** from the inside of the front side frame **6**.

[0105] In addition, as described above, the reinforcing plates **14c** can be disposed inside the front side frame **6** by being inserted integrally with the pipe nut main body **14a** from below the front side frame **6**. Therefore, even if the joining flange **14b** is provided at the base end of the pipe nut main body **14a**, the joining flange **14b** does not hinder the assemblability of the reinforcing plates **14c**. Then, by providing the joining flange **14b**, the joining flange **14b** can abut on the outer surface of the front side frame **6** to restrict the movement of the pipe nut main body **14a** in the longitudinal direction. Furthermore, by joining the joining flange **14b** to the outer surface of the front side frame **6** by spot welding, the fixing structure for the pipe nut **14** can ensure the attachment rigidity of the pipe nut **14** also from the outside of the front side frame **6**.

[0106] Thus, the fixing structure for the pipe nut **14** can ensure the attachment rigidity by welding from the outside of the front side frame **6**, in addition to welding from the inside of the front side frame **6**. Therefore, the fixing structure for the pipe nut **14** can increase the attachment rigidity with respect to the front side frame **6**.

[0107] In addition, the front side frame **6** has the pair of upper and lower inner flanges **12d** protruding from the outer surface of the front side frame **6**. When any one of the pair of rotation restricting portions **14h** abuts on the inner flange **12d** extending from the lower wall **12c**, each of the reinforcing plates **14c** extends in the direction orthogonal to the longitudinal direction of the front side frame **6**. That is, by simply causing any one of the rotation restricting portions **14h** to abut on the inner flange **12d**, each of the reinforcing plates **14c** can be disposed at an appropriate position in contact with the inner surface of the front side frame **6**. This configuration improves workability in arc welding of the reinforcing plates **14c** to the inner surface of the front side frame **6**.

[0108] Furthermore, as described above, each of the reinforcing plates **14c** extends in the direction orthogonal to the longitudinal direction inside the front side frame **6**. Thus, each of the reinforcing plates **14c** can accurately reinforce the front side frame **6**. Therefore, the fixing structure for the pipe nut **14** can increase the attachment rigidity of the pipe nut **14** with respect to the front side frame **6** and the rigidity of the front side frame **6**.

[0109] Further, each of the reinforcing plates **14c** has a flat plate shape. With such a shape, it is possible to insert each of the reinforcing plates **14c** into the front side frame **6** just by providing each of the slits **12f** corresponding to the thickness of each of the reinforcing plates **14c** in the lower wall **12c**. Thus, it is possible to minimize the number of holes provided in each of the reinforcing plates **14c**. Therefore, even with a structure in which the pipe nut main body **14a** including the

reinforcing plates **14c** is inserted into the front side frame **6**, it is possible to minimize a decrease in rigidity of the lower wall **12c**.

[0110] FIG. 7 illustrates a second embodiment of the disclosure. Components similar to those in the first embodiment are denoted by similar reference numerals, and description thereof is simplified or omitted.

[0111] The pipe nut main body **14a** of the first embodiment is disposed inside the front side frame **6**.

[0112] On the other hand, the pipe nut main body **14a** of the present embodiment penetrates the upper wall **12b** of the front side frame **6** and protrudes upward. That is, a distal end of the pipe nut main body **14a** of the present embodiment is different in that it is extended to the distal end side of the pipe nut main body **14a** of the first embodiment.

[0113] Therefore, the upper wall **12b** of the present embodiment is provided with an upper insertion hole **12g** through which the distal end of the pipe nut main body **14a** is inserted.

[0114] The pipe nut main body **14a** inserted into the upper insertion hole **12g** is rotatable around the central axis O of the pipe nut main body **14a**. Therefore, the diameter of the upper insertion hole **12g** is set slightly larger than the outer diameter of the pipe nut main body **14a**.

[0115] Then, the distal end side of the pipe nut main body **14a** inserted into the upper insertion hole **12g** can be joined to the outer surface of the upper wall **12b** by arc welding in a state of being inserted into the upper insertion hole **12g**.

[0116] Thus, the fixing structure of the present embodiment can generate a higher drag with respect to loads in the prying direction and the longitudinal direction input to the pipe nut main body **14a** via the suspension cross member **9** during traveling, as compared with the first embodiment.

Therefore, the fixing structure of the present embodiment can further increase the attachment rigidity of the pipe nut **14** with respect to the front side frame **6**.

[0117] Since the other configurations are similar to those of the first embodiment, the description thereof will be omitted.

[0118] FIGS. 8 and 9 illustrate a third embodiment of the disclosure. Components similar to those in the first and second embodiments are denoted by similar reference numerals, and description thereof is simplified or omitted.

[0119] The present embodiment is different from the first embodiment in that the front side frame **6** of the present embodiment is additionally provided with a joining portion **12k** that is in surface contact with any one of the pair of reinforcing plates **14c**.

[0120] The joining portion **12k** protrudes inward from the inner wall **12a** of the front side frame **6**.

[0121] For example, as illustrated in FIG. 9, the joining portion **12k** is provided near the center of the inner wall **12a** in the vehicle body up-down direction so as to protrude from the inner wall **12a** to the inside of the inner frame **12**.

[0122] The joining portion **12k** can be formed by a part of the plate material constituting the inner wall **12a**. That is, the joining portion **12k** is formed, for example, by providing a substantially U-shaped cut in a part of the plate material of the inner wall **12a**. By bending the base portion of the formed joining portion **12k** toward the inside of the inner frame **12** at a substantially right angle, the joining portion **12k** protrudes into the front side frame **6**. Thus, any one surface of the reinforcing plates **14c** rotated around the central axis O of the pipe nut main body **14a** can be brought into surface contact with the joining portion **12k**.

[0123] When the reinforcing plate **14c** comes into surface contact with the joining portion **12k**, the rotation of the pipe nut **14** is restricted. In one embodiment, the joining portion **12k** serves as a “second protrusion” that abuts on the reinforcing plate **14c** to restrict the rotation of the pipe nut **14**.

[0124] Further, when the reinforcing plate **14c** abuts on the joining portion **12k** by surface contact, each of the reinforcing plates **14c** extends in the direction orthogonal to the longitudinal direction of the inner frame **12**.

[0125] Then, in the abutted state, the reinforcing plate **14c** is joined to the joining portion **12k** by

spot welding.

[0126] For example, in the present embodiment, spot welding is performed at two positions in the direction of the central axis O of the pipe nut main body **14a** in a state where the reinforcing plate **14c** abuts on the joining portion **12k**.

[0127] With such a configuration, in the fixing structure of the present embodiment, the reinforcing plate **14c** can be joined to the front side frame **6** by spot welding. Thus, the disclosure is not limited to the arc welding described above, and the reinforcing plate **14c** can be joined to the front side frame **6** by spot welding. Therefore, the fixing structure of the present embodiment can further improve the assemblability of the pipe nut **14** with respect to the front side frame **6** depending on the vehicle type to be applied. Note that, in the present embodiment in which the reinforcing plate **14c** can be extended in the direction orthogonal to the longitudinal direction of the inner frame **12** just by bringing the reinforcing plate **14c** into surface contact with the joining portion **12k**, the pair of rotation restricting portions **14h** can be omitted.

[0128] Since the other configurations are similar to those of the first embodiment, the description thereof will be omitted.

[0129] Note that the disclosure described in the above embodiments (first to third embodiments) is not limited to those embodiments, and various modifications can be made in the implementation stage without departing from the gist thereof.

[0130] For example, the fixing structure of the pipe nut **14** described in each of the above embodiments (first to third embodiments) can also be used for a fastening portion between a vehicle body and a seat, a fastening portion between a vehicle body and a battery case mounted on an electric vehicle, or the like.

[0131] Furthermore, the above-described embodiments include inventions at various stages, and various inventions can be extracted by appropriately combining disclosed constituent elements.

[0132] Further, for example, when the problem to be solved in the disclosure can be solved and the described effect can be obtained even if some constituent elements are deleted from all the constituent elements described in the above embodiments, the configuration from which the constituent elements are deleted can be extracted as an invention.

Claims

1. A fixing structure for a pipe nut that fixes a pipe nut main body to a vehicle body frame having a hollow cross section, the fixing structure comprising: an insertion hole provided in the vehicle body frame and through which the pipe nut main body is inserted in such a manner that the pipe nut main body is rotatable around a central axis of the pipe nut main body; a slit provided in the vehicle body frame and extending from the insertion hole; a first stopper that is provided at a base end of the pipe nut main body and abuts on an outer surface of the vehicle body frame to restrict movement of the pipe nut main body in a longitudinal direction; and a second stopper that extends from a side surface of the pipe nut main body, passes through the slit, and is disposed inside the vehicle body frame, wherein the second stopper is joined to an inner surface of the vehicle body frame at a position rotated around the central axis from the slit.
2. The fixing structure for a pipe nut according to claim 1, wherein the vehicle body frame includes a first protrusion protruding from the outer surface, and when the first stopper abuts on the first protrusion, the second stopper extends in a direction orthogonal to a longitudinal direction of the vehicle body frame.
3. The fixing structure for a pipe nut according to claim 1, wherein the vehicle body frame includes a second protrusion protruding toward an inside of the vehicle body frame, and when the second stopper abuts on the second protrusion, the second stopper extends in a direction orthogonal to a longitudinal direction of the vehicle body frame.
4. The fixing structure for a pipe nut according to claim 1, wherein a distal end of the pipe nut main

body protrudes through the vehicle body frame.

5. The fixing structure for a pipe nut according to claim 1, wherein the first stopper is joined to the outer surface of the vehicle body frame by spot welding.

6. The fixing structure for a pipe nut according to claim 2, wherein the first stopper is joined to the outer surface of the vehicle body frame by spot welding.

7. The fixing structure for a pipe nut according to claim 3, wherein the first stopper is joined to the outer surface of the vehicle body frame by spot welding.

8. The fixing structure for a pipe nut according to claim 4, wherein the first stopper is joined to the outer surface of the vehicle body frame by spot welding.
